GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Diploma Engineering

Level: Diploma

Branch : Computer Engineering

Subject Code : DI05007011

Subject Name: Computer Hardware Architecture and System Maintenance

W.E. F. Academic Year:	2026-27
Semester:	5th
Category of the Course:	PCC

Prerequisite:	Students should have basic knowledge of: ● Fundamentals of computers and operating systems ● Basic electronics concepts (power supply, voltage, connectors) ● Computer peripherals and their basic usage ● Safe handling of electrical equipment
Rationale:	This course provides essential knowledge of computer hardware architecture, including internal components, processor organization, memory hierarchy, storage devices, and input/output peripherals. It enables students to understand system structure, performance factors, and communication between hardware components. The course also emphasizes practical skills in PC assembly, troubleshooting, and preventive maintenance. These competencies are essential for careers in hardware servicing, technical support, system maintenance, and IT infrastructure management.

Course Outcome:

After Completion of the Course, the student will be able to:

No	Course Outcomes	RBT Level
CO1	Identify and elaborate components of PC/laptop and overall system architecture.	R, U
CO2	Examine the processor, motherboard, and system buses.	R, U
CO3	Identify memory hierarchy and differentiate various memory types.	R, U
CO4	Demonstrate the working principles of input/output devices and peripherals.	A
CO5	Diagnose hardware faults and perform preventive maintenance of computer systems.	A

**Revised Bloom's Taxonomy (RBT)*

Teaching and Examination Scheme:

Teaching Scheme (in Hours)			Total Credits L+T+ (PR/2)	Assessment Pattern and Marks				Total Marks
				Theory		Tutorial / Practical		
L	T	PR	C	ESE (E)	PA(M)	PA(I)	ESE(V)	
3	0	2	4	70	30	20	30	150

Course Content:

Unit No.	Content	No. of Hours	% of Weightage
1.	**Core Components of PC/Laptop** • Introduction of computer systems and PC development • Types of computers - Micro - Mini - Mainframe - Supercomputer - Workstation - Cloud - Embedded Systems • PC block diagram and system components - CPU (ALU & CU) - Motherboard - RAM - ROM - Storage Device (HDD/SSD) - Keyboard - Mouse - Monitor (LCD/LED) - SMPS • Hardware, Software, Firmware concepts • BIOS, UEFI and CMOS concept • Components of a laptop - CPU - Memory(RAM)	**07**	**16**

	- Expansion Slots - Power supply(Battery) - Video display controller - Display - Removable media drives - Internal storage - Input - Ports - Cooling (Fan/Heat sink) - Speakers & Webcam - Wireless Module (Wi-Fi & Bluetooth) • Front & back panel overview - Ports ● USB Port ● HDMI Port ● RJ45 (Ethernet) Port ● Audio Port (3.5 mm Jack) ● VGA Port ● Thunderbolt Port - Connectors ● USB Connector ● HDMI Connector ● RJ45 Connector ● Audio Jack Connector ● VGA Connector ● Thunderbolt Connector • Prepare the latest PC/laptop specifications - Processor - RAM - Storage - Graphics Unit (GPU) - Display - Ports - Connectivity - Operating system. **Sample Configuration of Laptop : (**Intel Core i5 (13th Generation, up to ~4.6 GHz, multi-core, 16 GB DDR4**,** 512 GB SSD, **GPU:** Integrated Intel UHD / Iris Xe Graphics, 15.6-inch / 16-inch Full HD or WUXGA (1920×1080 or 1920×1200), IPS		

	panel, USB 3.2, USB-C, HDMI, Audio Jack, Charging Port, Wi-Fi 6 / Wi-Fi 6E, Bluetooth 5.x, Windows 11 Home / Pro (64-bit))		
2.	**Processor Architecture & Motherboard** • Introduction to Microprocessor • Overview and Features of 8085 Microprocessor • Block Diagram of 8085 Architecture: ● Registers (Accumulator, General Purpose, PC, SP) ● ALU and Control Unit ● Buses (Address, Data, Control) • Pin Diagram of 8085 Microprocessor • Distinguish between RISC and CISC Architecture • Multi-core processor-concept & advantages - Dual Core - Quad Core • Motherboard introduction - Block Diagram - Components - Layout • Motherboard types - AT - ATX - Micro-ATX - Mini-ITX • Chipset architecture - Northbridge - Southbridge • Expansion buses & slots - ISA - PCI - PCI-E - PCI-X - AGP - USB - VESA local bus	**11**	**25**

	• SMPS working & connectors • Storage interfaces - SATA - NVMe - RAID		
3.	**Memory Organization** • Memory hierarchy & classification • Primary storage(RAM & ROM) • RAM types - SRAM - DRAM - DDR generations • ROM types - PROM - EPROM - EEPROM - Flash • Secondary Storage • HDD(Hard Disk Drive) - HDD structure and working - Disk performance parameters - Seek time - Rotational latency - Disk access time - Data transfer rate • Solid State Drive (SSD) - Basic concept - Types and Working Principle • HDD vs SSD • DVD, USB • Cache memory levels (L1, L2 and L3) & significance • Associative memory concept • Virtual memory concept - Paging	**08**	**18**

	- segmentation		
4.	**Input, Output & Peripheral Devices** • Types of Keyboard(List out Only) • Keyboard- Working principle • Mechanical Keyboard - Construction - Operation • Membrane Keyboard - Construction - Operation • Wireless Keyboard - Construction - Operation • Types of Mouse (List out Only) • Mouse- Working principle • Optical mouse - Construction - Operation • Wireless (Bluetooth) Mouse - Construction - Operation • Types of Scanner (List out Only) • Scanner- Working principle • Flatbed Scanner - Construction - Operation • Sheet-fed Scanner - Construction -Operation • Types of Monitor (List out Only) • LCD	**12**	**25**

	- Working Principle - Construction -Operation • LED - Working Principle - Construction -Operation • Types of Printer (List out Only) • Laser Printer - Working Principle - Construction -Operation -Application • Inkjet Printer - Working Principle - Construction -Operation -Application • Thermal Printer - Working Principle - Construction -Operation -Application		
5.	**PC Troubleshooting & Preventive Maintenance** • POST - Functions - IPL Hardware - Test Sequences - Error messages with codes • Visual inspection techniques • Layman checks for PC Troubleshooting • Systematic Troubleshooting Methodology	**07**	**16**

	- Motherboard troubleshooting - Keyboard troubleshooting - Mouse troubleshooting - Printer troubleshooting - HDD troubleshooting - Monitor troubleshooting - RAM troubleshooting - SMPS troubleshooting • Preventive maintenance - Safety practices steps - Troubleshooting Tools - Digital Multimeter - Oscilloscope - Logic Analyzer - Logic Probe - Logic Pulser - Current Tracer • Diagnostic Software - Memtest - CCleaner - HDD scan - defraggler - CPU-Z - CrystalDiskInfo • PC Assembling and disassembling		
	Total	45	100

Suggested Specification Table with Marks (Theory):

Distribution of Theory Marks (in %)					
R Level	**U Level**	**A Level**	**N Level**	**E Level**	**C Level**
30	40	30	**0**	**0**	**0**

Where R: Remember; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create (as per Revised Bloom's Taxonomy)

References/Suggested Learning Resources:

(a) Books:

a) Craig Zacker & John Rourke, "PC Hardware: The Complete Reference", McGraw-Hill Education.

b) Scott Mueller, "Upgrading and Repairing PCs", Que Publishing, Latest Edition, ISBN: 978-0789756107.

c) William Stallings, "Computer Organization and Architecture", Pearson Education, Latest Edition, ISBN: 978-9332582705.

d) K.L. James, "Computer Hardware Installation & Interfacing", PHI Learning, New Delhi.

e) Stephen J. Bigelow, "Troubleshooting, Maintaining & Repairing PCs", McGraw-Hill Education.

f) Mark Minasi, "The Complete PC Upgrade and Maintenance Guide", Wiley India.

g) Mike Meyers, "CompTIA A+ Certification All-in-One Exam Guide", McGraw-Hill Education, ISBN: 978-1260454038.

h) Morris Mano, "Computer System Architecture", Pearson Education.

(b) Open-source software and website:

a) https://www.computerhope.com
b) https://www.ifixit.com
c) https://www.howstuffworks.com
d) https://www.intel.com
e) https://www.amd.com
f) https://www.crucial.com/support
g) https://www.nptel.ac.in
h) https://www.tutorialspoint.com

Suggested Course Practical List:

1. Identify internal components of a desktop computer and observe their functions.
2. Study & Comparison of CISC and RISC Architecture
3. Identify motherboard components including CPU socket, RAM slots, chipset, expansion slots and connectors.
4. Observe and list the functions of front panel and back panel ports of a PC.
5. Study SMPS connectors and measure output voltages using a digital multimeter.
6. Install and remove RAM modules and verify detection in BIOS/UEFI.
7. Install HDD/SSD and connect SATA/NVMe interface cables.
8. Configure BIOS/UEFI settings and observe hardware detection parameters.
9. Identify different types of memory modules and storage devices.
10. Connect and test input devices such as keyboard, mouse and scanner.
11. Install and configure a printer and perform a test print.
12. Identify monitor ports and configure display settings in the operating system.
13. Perform POST and interpret beep codes and startup error messages.
14. Troubleshoot common hardware faults such as no display, boot failure, and RAM errors.
15. Assemble and disassemble a PC cabinet following safety and ESD precautions.

List of Laboratory/Learning Resources Required:

1. Desktop computers with basic configuration and Windows/Linux operating systems.
2. Computer hardware components for practice (motherboard, CPU, RAM, SMPS, HDD/SSD).
3. PC hardware tool kit (screwdriver set, anti-static wrist strap, cleaning brush, thermal paste).
4. Digital multimeter for power supply testing and diagnostics.
5. Diagnostic and utility software (CPU-Z, HWMonitor, MemTest86, disk diagnostic tools).
6. Input devices such as keyboard, mouse, and scanner.
7. Output devices such as LED/LCD monitors and printers.
8. Storage devices, including pen drives, external HDD/SSD, and optical media.
9. Network cables, connectors, and internet connectivity.
10. UPS or a power backup system for safe operation.

Suggested Activities for Students:

Other than classroom and laboratory learning, the following co-curricular activities are suggested to accelerate attainment of course outcomes. Students should perform these activities in groups and prepare brief reports with supporting evidence for their academic portfolio and placement readiness.

a) Conduct group discussions on **types of memory and memory hierarchy**.
b) Undertakes micro-projects related to PC assembly, troubleshooting, or hardware comparison.
c) Deliver seminars on the **latest processor architectures** (e.g., Intel Core series, AMD Ryzen).
d) Prepare charts and comparison posters on hardware components such as HDD vs SSD, RAM types, or motherboard form factors.
e) Collect and study older storage devices (floppy disks, magnetic tapes, CDs/DVDs) to understand
technological evolution.
f) Demonstrate PC assembly and disassembly procedures.
g) Prepare preventive maintenance checklists for computer labs.
h) Visit a computer service center or IT lab to observe hardware maintenance practices.

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